Total No. of Printed Pages:02

SUBJECT CODE NO: RNP-8015
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Civil Engg.) (NEP) Sem - III
Examination May/June-2026
Strength of Material

[Time: 1:30 Hours] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1. Part A: Question No 1 is compulsory.
2. Attempt any TWO questions from remaining Part B: Q.2 to Q.5.

PART – A

Q1 Solve any 5 following questions **10**

a) Define strain and write its mathematical expression.
b) What is modulus of rigidity?
c) Define thermal stress.
d) What is point of contra flexure?
e) Define polar moment of inertia.
f) What is effective length of column?
g) State Euler's theory of columns.

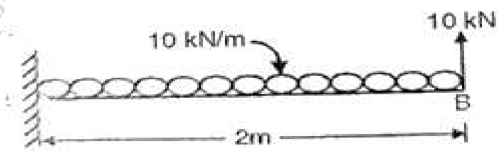
PART – B

Q.2 Solve any TWO from the following

10

1) A compound bar consists of a steel rod and a brass rod rigidly connected together. The system is subjected to an axial load of 120 kN. Determine the stresses developed in each material.
Take: Area of steel rod= 900 mm^2, Area of brass rod= 1200 mm^2,
$E_s = 200\,GPa, E_b = 100\,GPa$

2) Draw SFD and BMD of given beam. **10**



3) T Section with flange ($320\,mm \times 80\,mm$) and web ($100\,mm \times 80\,mm$), is used as a cantilever beam of 3.6 m span subjected to a u.d.l. of intensity 180 kN/m over its full span. Determine the maximum stresses in the beam. **10**

4) Analyse the given truss by joint method. Also draw the nature of forces in each member. **10**